

SUMMARY

Senior Unity and AI gameplay engineer available for contract engagements, with 5+ years across **shipped commercial games**, **enterprise R&D**, and **freelance product delivery**. Specializes in **VR/WebGL/PC character systems**, **LLM/TTS product integration**, and production architecture. At **Squirle Games / SecretWaifu**, shipped cross-repo Unity and web/backend work spanning custom FSM/blackboard architecture, WebGL launch/auth, streamed chat/TTS, PostgreSQL migration tooling, and agentic AI-assisted development workflows.

SKILLS

- **Unity / VR / WebGL:** Unity 6, C#, OpenXR, XR Interaction Toolkit, VRM avatars, UI Toolkit, Addressables, WebGL, PC/VR runtime switching, profiling and optimization.
- **AI / LLM / TTS:** Multi-provider LLM integration, SSE/WebSocket streaming, provider chat templates, structured JSON/GBNF outputs, TTS/STT, lip sync, prompt diagnostics.
- **Architecture:** Awaitable async flows, custom FSMs, StateProvider/Blackboard patterns, platform strategies, persistence/cache design, runtime logging, production debugging.
- **Backend & Product Infrastructure:** TypeScript, Node/Express, Next.js, Prisma, PostgreSQL/SQLite migration tooling, auth, quota/subscription systems, WebSocket/SSE APIs.
- **Agentic AI Engineering:** Codex/Claude-assisted architecture, implementation, code review, research, migration planning, Unity MCP automation, custom repo skills, verification guardrails.
- **Gameplay / Graphics / Networking:** Interaction and physics systems, Unity Netcode, Mirror Networking, HLSL/ShaderLab, URP, NiloToon, FinalIK.

EXPERIENCE

- **Squirle Games / SecretWaifu** Remote
Senior Unity & Full-Stack AI Game Developer (Contract) December 2025 – Present
 - Own cross-platform AI character product architecture across the Unity client and SecretWaifu web/backend, shipping PC/WebGL/VR releases with VRM avatars, OpenXR UI, secure launch/auth, quota/subscription flows, and backend session state.
 - Rebuilt the Unity runtime around an Awaitable-based FSM framework, StateProvider/Blackboard coordination, platform strategies, API/SSE transport, structured logging, and editor diagnostics; delivered wrist menu, spatial keyboard, PC/VR switching, expression/animation pipelines, and FinalIK/NiloToon physical-response tuning.
 - Integrated the real-time LLM/TTS/STT pipeline: multi-provider chat templates, backend structured-helper generation, streaming visible chat, turn-scoped TTS WebSockets, WebGL chunked audio playback, lip sync, typewriter timing, and animation/gesture synchronization.
 - Hardened the TypeScript/Prisma backend with WebGL auth bridge security, membership game-access gates, durable quota/counter ledgers, Patreon entitlement fixes, landing/marketing automation, story/default-model refactors, and admin repair tooling.
 - Built PostgreSQL migration and cutover tooling for the website backend, including SQLite import rehearsal, row conversion, raw-SQL ownership, fingerprint/count validation, and cleanup of SQLite-specific runtime paths.
 - Used agentic AI as a production workflow: Codex/Claude, Unity MCP, and custom repo skills for research-backed planning, architecture reviews, migration audits, code generation, guard checks, and cross-repo implementation tracking.

- Capgemini Engineering** Bologna – Remote
Unity Software Engineer (R&D) October 2023 – February 2026
 - Refactored a custom Obi-physics simulator integration, resolving CPU bottlenecks and raising target-hardware framerate from **10 FPS to stable 60 FPS**.
 - Engineered Unity Addressables content delivery to decouple assets from the binary, reduce build size, and support secure on-demand remote loading.
 - Built WebGL/VR museum and showcase applications for a major telecom client, including an in-app editor, visitor experience, and real-time multiplayer via Unity Netcode.
 - Ported high-fidelity assets and interaction models across WebGL, mobile, and VR constraints; optimized serialization/save data for browser heap limits.
 - Created custom node-based editor tooling and CloudXR integrations to reduce scenario setup time from hours to minutes and support low-latency streaming.
- Gamma Interactive** Remote
Game Programmer March 2021 – September 2023
 - Contributed to *Alaloth: Champions of the Four Kingdoms* (Steam Early Access), shipping a complex Action-RPG with over **10k wishlists**.
 - Developed in-engine tools for asset validation and world management, reducing manual production operations from hours to seconds.
 - Implemented gameplay loops, UI architecture, inventory/skill systems, narrative events, and production bug fixes in close collaboration with QA.

SELECTED GAME PROJECTS

- **Alaloth: Champions of the Four Kingdoms** (Steam / PC, Game Programmer, 2021–2023): shipped ARPG features for UI, narrative events, and modular gameplay systems. [Steam]
- **Apocalypse Rangers** (PC / Online, Gameplay & Network Programmer, 2020–2021): built Mirror client-server multiplayer and custom HLSL toon/terrain rendering; **New Game Designer 2020, 3rd Place Online**. [Trailer]
- **Hell for One** (PC / RTS Hybrid, AI & Gameplay Programmer, 2019–2020): designed agent AI, steering/state-machine behavior, and large-battle spawning; **New Game Designer 2020, 1st Place Offline**. [Itch.io]
- **RuNeglect** (Microsoft HoloLens, Sole Developer, 2018–2019): built rehabilitative AR thesis app using HoloLens Spatial Mapping and object pooling for room-adaptive gameplay.

EDUCATION

- **Università degli Studi di Milano**, Master's Coursework in Computer Science (Game Dev Path) 2019–2021
Advanced Graphics, Real-Time Rendering, and Artificial Intelligence.
- **Università degli Studi di Milano**, B.S. in Computer Science 2019
Thesis: “Development of an AR application for rehabilitation in hemispatial neglect” (Microsoft HoloLens).

LANGUAGES

- **Italian:** Native Speaker
- **English:** Professional Working Proficiency